

Rhenium

Re (Re I). Strongest classified transitions below 2 μm (NIST): 67 levels, 90 lines.

Re⁺ ionisation limit: 63181.60 cm⁻¹ = 7.83352 eV (158.274 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U|e|J||J\rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

